Hardware Documentation: Three-Board Robot Electronics

1. Hardware Overview

Board	Descriptive name	Verified function	Key parts
1	H-Bridge Motor Driver	Full H-bridge (two half-bridges) for one DC motor, 12 V supply	2x IR2104PBF, 4x IRF1404PBF
2	ESP32 Controller / Peripheral Interface	Hosts an ESP32 dev module and breaks out motor-control signals, encoders, servos, IR sensors, ultrasonic sensor and an I2C IMU	ESP32-WROOM-32 dev module
3	5 V Buck Regulator	Steps battery voltage down to 5 V; also passes battery voltage to a DRIVER connector	XL4015E1

Architecture summary (Interpretation): Board 3 serves as the system power source. Its DRIVER output is intended to supply the motor driver (Board 1), and its 5 V header is intended to supply the controller board (Board 2), based on connector and net names. The control interface between Board 2 and Board 1 is described in Section 6.

2. Hardware Architecture

Power flow (Interpretation):

- The battery connects to Board 3 BATT.
- Board 3 DRIVER connector is electrically paralleled with the buck input (Confirmed on schematic), so it is a battery pass-through.

- Board 3 5 V header feeds sensors and servos via Board 2 +5V connector.
- Board 1 receives its 12 V (labelled +12V) on U11.

Signal flow (Confirmed per board):

- Board 1 accepts PWM1, EN, PWM2 and GND on a 4-pin header (U4).
- Board 2 outputs the signals listed in Section 4.4 (servo PWM, motor-control signals M1/M2 with PWM4/PWM5, I2C, encoder inputs, IR inputs, ultrasonic TRIG/ECHO).
- Net-name caution: Board 1 nets PWM1/PWM2 and Board 2 nets PWM1/PWM2 are in separate schematics. On Board 2 they drive SERVO1 and SERVO2. Identical names must not be treated as a connection.

3. Board 1: H-Bridge Motor Driver

3.1 Overview

A discrete full-bridge driver: two IR2104PBF half-bridge gate drivers (U1, U2) each drive one high-side and one low-side IRF1404PBF N-MOSFET. U1 with Q1/Q2 forms the M+ leg; U2 with Q3/Q4 forms the M- leg (Confirmed from schematic).

3.2 Functional blocks

Block	Components	Function
Gate drivers	U1, U2 (IR2104PBF)	Half-bridge drivers: IN sets the leg state; SD# pin tied to EN net
Power stage	Q1, Q2 (leg M+), Q3, Q4 (leg M-)	IRF1404PBF N-channel MOSFETs, TO-220
Bootstrap	D1, D6 (UF4007) and C3, C5 (4.7 uF)	Supply for floating high-side drive (VB to VS)
Gate networks	R3 to R6 (10 ohm) with D2 to D5 (1N4148) in parallel	Gate resistor with parallel diode; diode appears to bypass the resistor for fast turn-off
Freewheel	D12 to D15 (HER308)	One diode across each MOSFET drain-source
Motor-side protection	D7 (P6KE18CA bidirectional TVS), R8 (10	TVS across motor terminals; RC snubber

Block	Components	Function
	ohm) + C7 (100 nF)	across motor terminals
Decoupling / bulk	C1, C6 (100 nF) at driver VCC; C4 (4.7 uF), C8 (470 uF) at input	Supply filtering
Pull-downs	R1, R2 (10 kohm) on U1 SD#; R7 (10 kohm) on U2 SD#	Hold SD# low when EN is not driven
Cooling	U6 to U9 (XSD1226-005-21 heat sinks)	One per MOSFET

3.3 Circuit operation (Interpretation, consistent with schematic)

- Each IR2104 produces complementary HO/LO from a single IN input, so the board needs one PWM signal per leg (PWM1 for the M+ leg, PWM2 for the M- leg), with the shoot-through dead time provided inside the IR2104.
- Each bootstrap capacitor charges through its UF4007 from +12V while the low-side MOSFET conducts. A leg therefore cannot hold 100% high-side duty indefinitely.
- Both drivers share one EN net on SD#. The schematic symbol labels the pin SD# (suggesting active-low), while the 10 kohm pull-downs hold it low. Published IR2104 datasheets describe SD as active-high shutdown; under that behavior, EN high would disable the drivers.
- Heat sinks U6 to U9 are connected to nets Drain1 to Drain4 (Confirmed on schematic). Two sit on +12V and two on the motor switching nodes.

3.4 Power

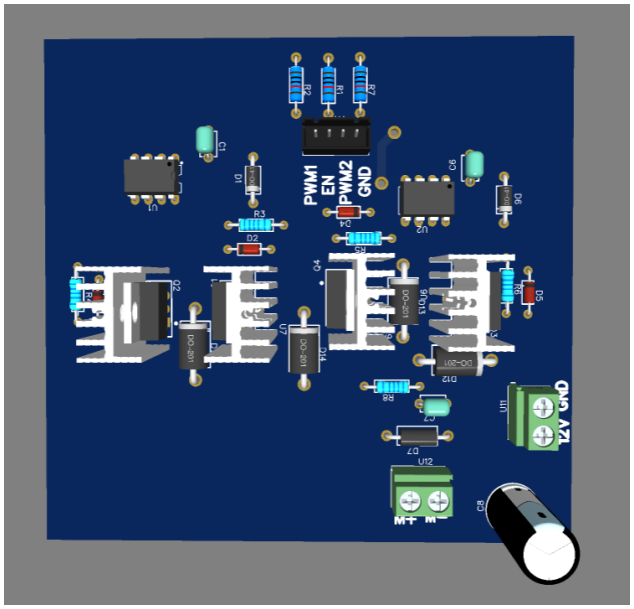
- Input: U11 (KF126-5.0-2P), labelled +12V and GND on the PCB silkscreen.
- Driver VCC: +12V directly (no on-board regulator). The IR2104 VCC operating range is approximately 10 to 20 V according to published specifications.
- No fuse, reverse-polarity protection or input TVS is present in the schematic or BOM.

3.5 Connectors

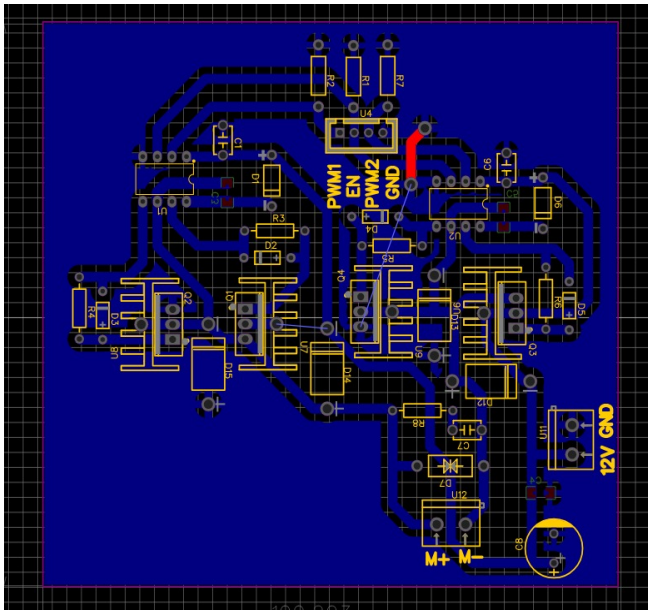
Ref	Type	Pins (Confirmed from schematic and silkscreen)
U4	B4B-XH-A 4-pin	1 PWM1, 2 EN, 3 PWM2, 4 GND
U11	KF126-5.0-2P	+12V, GND (pin order)
U12	KF126-5.0-2P	M+, M- (motor)

3.6 Electrical characteristics

Published specifications identify the IRF1404PBF as a 40 V, low-RDS(on) device with a large gate charge (approximately 160 nC). The IR2104 output drive current is on the order of 130 mA source / 270 mA sink. Together with the 10 ohm gate resistors, these values imply switching times in the low-microsecond range, so PWM frequency selection should account for switching losses. No current-sense element is present on the board.



4.



Board 2: ESP32 Controller / Peripheral Interface

4.1 Overview

A carrier board for an ESP32 development module (symbol: ESP32-Wroom-32; BOM: ASTE83_AZ-DELIVERY-ESP32-WROOM-32; footprint: AI-THINKER_NODEMCU-32SLUA). It contains no regulator and no power stage. It

routes MCU pins to field connectors and provides I2C pull-ups and one decoupling capacitor.

4.2 Functional blocks

Block	Components	Function
Controller	U1	ESP32 dev module, powered from +5V on its Vin pin
Power input / decoupling	+5V connector (KF127-5.08-2P), C1 (100 nF)	5 V entry and local decoupling
I2C pull-ups	R1 (1k, SCL), R2 (1k, SDA)	Pull-ups to +3V3
Motor-control connectors	M1, M2 (KF127-5.0-3P)	Three signals each, see 4.4
Encoders	ENCODER1, ENCODER2 (KF2EDGR-3.81-4P)	+5V, GND, channel A, channel B
Servos	SERVO1 to SERVO3 (KF127-5.0-3P)	+5V, PWM signal, GND
Sensors	IR1, IR2 (1x3 header), ULTRASONIC (1x4), IMU (1x4)	Peripheral headers

4.3 Power

- +5V from the external connector feeds the module Vin pin and all peripheral connectors. +3V3 comes from the dev module 3.3 V pin and is used only for the I2C pull-ups.
- No protection or fuse is present. Servo and sensor current is drawn directly through the 5 V connector and board copper.

4.4 Connector pin assignments

Pin assignments are based on the schematic and PCB silkscreen.

Connector	Pin 1	Pin 2	Pin 3	Pin 4
ENCODER1	+5V	GND	ChannelA 1	ChannelB 1
ENCODER2	+5V	GND	ChannelA 2	ChannelB 2

Connector	Pin 1	Pin 2	Pin 3	Pin 4
IR1 / IR2	+5V	IR1 / IR2 signal	GND	n/a
ULTRASONIC	+5V	ECHO	TRIG	GND
IMU	+5V	GND	SCL	SDA
SERVO1 / 2 / 3	+5V	PWM1 / PWM2 / PWM3	GND	n/a
M1	M1+	PWM4	M1-	n/a
M2	M2+	PWM5	M2-	n/a

M1 and M2 do not carry motor power. They are 3-signal connectors whose function (for example direction inputs plus a speed PWM to an external driver) is an Interpretation only.

4.5 ESP32 pin mapping (read from schematic)

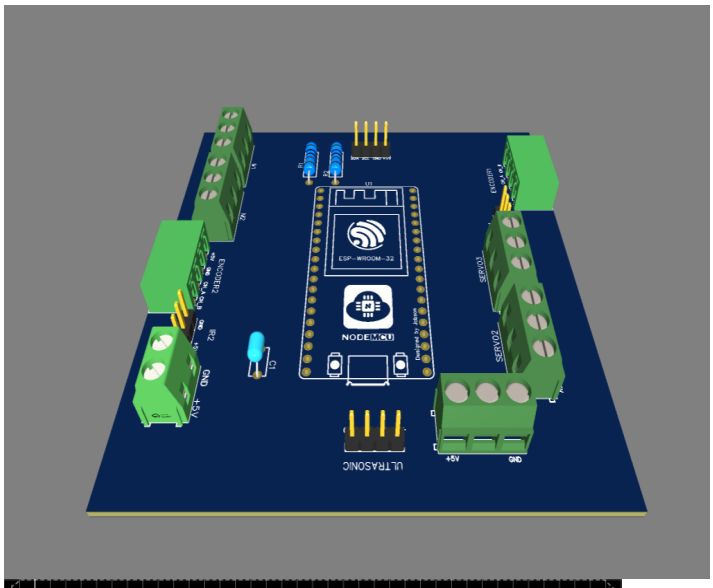
Signal	MCU pin	Notes
PWM1, PWM2, PWM3	CLK, SD0, SD1	See issue C1 below
ECHO, TRIG	SD2, SD3	See issue C1 below
PWM4	SVP (GPIO36)	Input-only pin, see C2
M1-	SVN (GPIO39)	Input-only pin
M1+	Label sits at the EN row	Schematic label adjacent to EN row
M2+ / PWM5 / M2-	P32 / P33 / P25	
ChannelA2 / ChannelB2	P26 / P27	
ChannelA1 / ChannelB1	P18 / P19	
IR2	P14	
IR1	P16 or P4	Label sits between the two rows
SCL / SDA	P22 / P21	
Unused	P34, P35, P12, P13, P0, P2, P5, P15, P17, P23, TX, RX	Marked no-connect

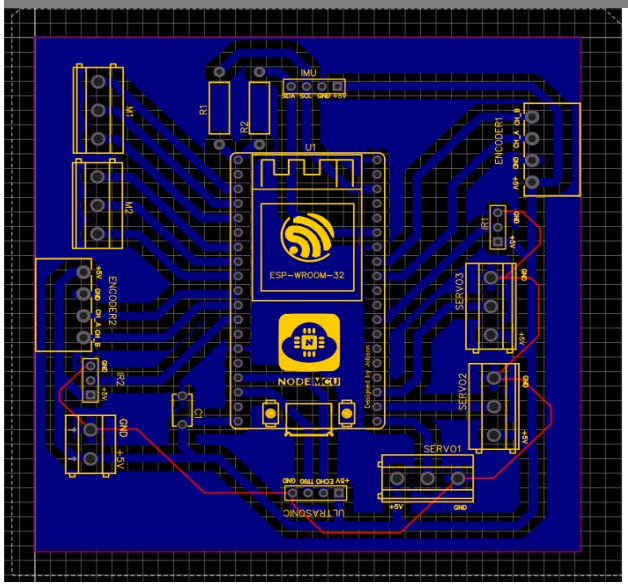
4.6 Findings specific to this board

- C1 (Confirmed on schematic, rule per Espressif documentation): The pins named CLK, SD0, SD1, SD2, SD3 correspond to GPIO6 to GPIO11, which connect to the internal SPI flash on ESP32-WROOM-32 modules and are not

available as general I/O. Five signals (PWM1 to PWM3, ECHO, TRIG) are assigned to them. This is likely to prevent normal operation. Highest priority issue.

- C2 (Interpretation): PWM4 is on GPIO36 and M1- on GPIO39, both input-only pins. An output PWM cannot be generated on GPIO36.
- C3 (Potential): ECHO, IR and encoder signals originate from 5 V peripherals and connect directly to 3.3 V GPIOs with no divider or level shifter in the schematic. Safe only if each sensor output is 3.3 V compatible.
- C4 (Potential): The +5V connector feeds Vin; if the dev module is also powered over USB, 5 V sources may contend depending on the module's Vin/USB isolation.
- C5 (Potential): Servo connector order is +5V / signal / GND. Standard hobby servo plugs are GND / V+ / signal, so a direct plug-in may be reversed.
- C6 (Confirmed discrepancy): Schematic symbol/BOM name is an AZ-Delivery module while the footprint is Ai-Thinker NodeMCU-32S. Header spacing and pin order must be checked against the physical module.





5. Board 3: 5 V Buck Regulator

5.1 Overview

A step-down converter built around the XL4015E1 (TO-263-5), with a 5 V output header and a battery input. A DRIVER connector is paralleled with the input.

5.2 Functional blocks

Block	Components	Function
Converter IC	U1 (XL4015E1)	Monolithic buck regulator

Block	Components	Function
Input filtering	C1 (220 uF), C4 (1 uF), C5 (1 uF)	Input capacitors at VIN
Power inductor	L (toroidal line choke, 15W8 footprint)	Energy storage, switch node to output
Catch diode	D1 (SR540)	Schottky freewheel diode from SW to GND
Output filtering	C7 (330 uF), C6 (1 uF)	Output capacitors
Feedback	R1 (10 kohm), R2 (3.3 kohm), C2 (33 nF)	Output divider with capacitor across R1
Connectors	BATT, DRIVER (KF301-5.0-2P), 5V header (HDR 1x6)	Input, battery pass-through, output

5.3 Circuit operation and output voltage

Using the published XL4015 feedback reference of approximately 1.25 V, the divider gives Vout = 1.25 × (1 + 10k / 3.3k) ≈ 5.04 V (Interpretation, calculated from BOM values and schematic topology). C2 acts as a feed-forward capacitor across the upper resistor.

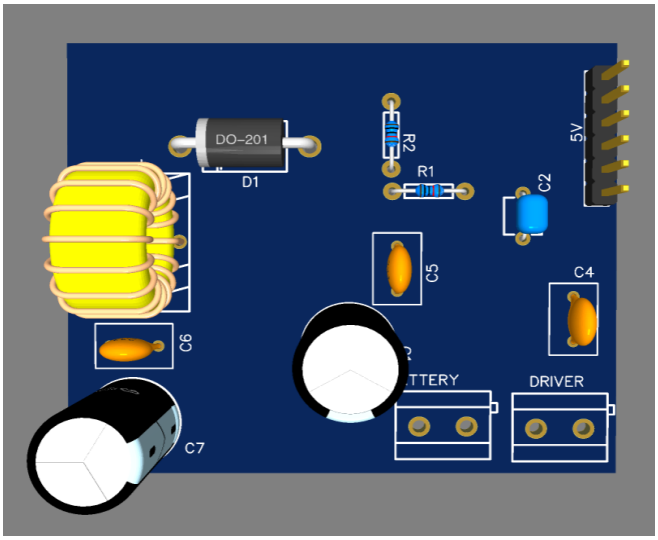
Schematic pin usage: pin 5 VIN, pins 3 and 6 SW (pin 6 is the tab), pin 2 FB, pin 1 GND, and pin 4 VC, which is wired to the VIN node in the schematic.

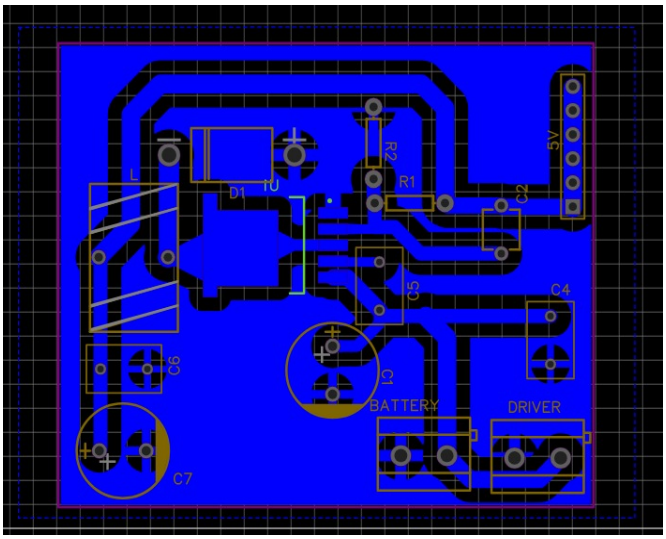
5.4 Electrical ratings

The XL4015 is commonly specified for approximately 8 to 36 V input and up to 5 A output; the SR540 is a 5 A, 40 V Schottky diode.

5.5 Connectors

Ref	Function
BATT (KF301-5.0-2P)	Battery input
DRIVER (KF301-5.0-2P)	Battery pass-through to the motor driver (name-based Interpretation)
5V (HDR 1x6)	Six 5 V output pins
GND header	Shown in schematic (1x6), not present in BOM or PCB





6. Hardware Integration

Link	Status	Evidence
Board 3 DRIVER to Board 1 U11 (12 V)	Interpretation	Connector names only; battery voltage must be 12 V-class
Board 3 5V header to Board 2 +5V connector	Interpretation	Net names only
Board 2 to Board 1 control	Not defined	Board 1 requires PWM1, EN, PWM2, GND; no Board 2 net with a matching function and no shared connector type

Link	Status	Evidence
Common ground	Required	Ground must be common between boards; not documented as a harness

Board 2 M1/M2 (M+, PWM, M−) uses a different signal set from Board 1 U4; the interfaces are not directly equivalent.

7. BOM Documentation

The BOMs are reproduced as supplied, preserving their component names, designators, footprints, and quantities.

7.1 Board 1: H-Bridge Motor Driver

ID	Name	Designator	Footprint	Qty
1	4.7uF	C5, C4, C3	C1206	3
2	470uF	C8	CAP-TH_BD10.0-P5.00-D0.6-FD	1
3	HER308	D12, D13, D14, D15	DO-201AD_BD5.3-L8.4-P12.50-D1.4-RD	4
4	KF126-5.0-2P	U11, U12	CONN-TH_P5.00_KF126-5.0-2P	2
5	100nF	C1, C7, C6	CAP-TH_L4.8-W3.2-P5.08-D1.2	3
6	XSD1226-005-21	U9, U8, U7, U6	HEATSINK-TH_L15.5-W10.5-XSD1226-005	4
7	B4B-XH-A-BK(LF)(SN)	U4	CONN-TH_B4B-XH-A-BK-LF-SN	1
8	IR2104PBF	U2, U1	DIP-8_L9.9-	2

ID	Name	Designator	Footprint	Qty
			W6.6-P2.54-LS7.62-BL	
9	10 ohm	R8, R6, R5, R4, R3	RES-TH_BD2.2-L6.5-P10.50-D0.6	5
10	10 kohm	R7, R2, R1	RES-TH_BD2.5-L6.5-P10.50-D0.6	3
11	IRF1404PBF	Q4, Q3, Q2, Q1	TO-220-3_L10.0-W4.5-P2.54-L	4
12	P6KE18CA_C713672	D7	DO-15_BD3.1-L6.7-P10.70-D0.9-BI	1
13	1N4148-C85057	D5, D4, D3, D2	DO-35_BD2.0-L4.2-P8.20-D0.5-RD	4
14	UF4007_C84067	D6, D1	DO-41_BD2.4-L4.7-P8.70-D0.9-FD	2

Cross-check: all schematic designators (C1 to C8, D1 to D7, D12 to D15, Q1 to Q4, R1 to R8, U1, U2, U4, U6 to U9, U11, U12) appear in the BOM with matching quantities.

7.2 Board 2: ESP32 Controller / Peripheral Interface

ID	Name	Designator	Footprint	Qty
1	KF2EDGR-3.81-4P	ENCODER1, ENCODER2	CONN-TH_4P-P3.81_KF2EDGR-3.81-4P	2
2	KF127-5.08-2P	+5V	CONN-TH_P5.08_KF127-5.08-2P	1
3	100nF	C1	CAP-TH_L5.5-	1

ID	Name	Designator	Footprint	Qty
			W3.2-P5.00-D1.0	
4	HDR-M-2.54_1x4	IMU, ULTRASONIC	HDR-M-2.54_1X4	2
5	HDR-M-2.54_1x3	IR1, IR2	HDR-M-2.54_1X3	2
6	KF127-5.0-3P	M1, M2, SERVO1, SERVO2, SERVO3	CONN-TH_3P-P5.00_KF127-5.0-3P	5
7	1k	R1, R2	R_AXIAL-0.5	2
8	ASTE83_AZ-DELIVERY-ESP32-WROOM-32	U1	AI-THINKER_NODEMCU-32SLUA	1

Cross-check: all schematic parts are present. The power connector uses designator +5V rather than a U-number, which is unusual but consistent between schematic and BOM.

7.3 Board 3: 5 V Buck Regulator

ID	Name	Designator	Footprint	Qty
1	HDR-M-2.54_1x6	5V	HDR-M-2.54_1X6	1
2	KF301-5.0-2P	BATT, DRIVER	CONN-TH_P5.00_KF301-5.0-2P	2
3	220uF	C1	CAP-TH_BD10.0-P5.00-D1.0-FD	1
4	33nF	C2	CAP-TH_L4.2-W3.8-P5.08-D0.6	1
5	1u	C4, C5, C6	RAD-0.2	3
6	330uF	C7	CAP-TH_BD10.0-P5.00-D1.0-FD	1
7	SR540_C129837	D1	DO-201AD_BD5.3-L8.4-P12.50-D1.4-RD	1
8	TOROIDALLINECHOKET15	L	IND-TH_L15.6-W9.3-P7.5-	1

ID	Name	Designator	Footprint	Qty
	W8		D0.6	
9	10k	R1	R_AXIAL-0.3	1
10	3.3k	R2	R_AXIAL-0.3	1
11	XL4015E1	U1	TO-263-5_L10.2-W8.9-P1.70-TL	1